

Controlling random seed stability in regression through bagging

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Abstract:

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1. Introduction

- Introduce random seed stability abstractly
- Discuss the recent increase in articles flagging random seed stability as an issue
- Connect the issue of random seed stability to that of algorithmic stability
- State the contributions of this work: bagging controls random seed stability

2. Random seed stability

- Discuss algorithmic stability
- Connect algorithmic stability to random seed stability
- Introduce the formal definition of random seed stability

Let $\mathbf{Z}_i = (X_i, Y_i)$, where $X_i \in \mathcal{X} \subseteq \mathbb{R}^d$ and, without loss of generality, $Y_i \in [0, 1]$. Let $\mathbf{Z} = \{\mathbf{Z}_i\}_{i=1}^n$ denote a fixed training dataset. Let $\mathcal{S} = \{1, \dots, M_{\text{int}}\}$ where M_{int} is the largest integer that can be represented on a machine; define $\mathcal{P}_{\mathcal{S}} := \text{Unif}(\mathcal{S})$. Let $\mathfrak{F}$ denote a generic regression algorithm and $\mathfrak{F}(\mathbf{Z}^{(n)}, s) = \hat{f}_n(\cdot; \mathbf{Z}^{(n)}, s) : \mathcal{X} \rightarrow [0, 1]$ be a realized regression function, where $s \in \mathcal{S}$ is a random seed. We define the algorithm function class

$$\mathcal{F}_n(\mathbf{Z}) = \{\mathfrak{F}(\mathbf{Z}, s) : s \in \mathcal{S}\}.$$

That is $\mathcal{F}_n(\mathbf{Z})$ is the finite (albeit very large) set of functions produced by training on the same dataset and architecture, but only varying the random seed.

Definition 2.1. An algorithm is random seed (ϵ, δ) -stable if for any training data $\mathbf{Z}$ and test data $x \in \mathcal{X}$

$$\mathbb{P}_{s, s' \sim \mathcal{P}_{\mathcal{S}}} \left(\left| \hat{f}_n(x; \mathbf{Z}, s) - \hat{f}_n(x; \mathbf{Z}, s') \right| \geq \epsilon \right) \leq \delta.$$

Assumption 1. Let $\{\mathbf{Z}^{(n)}\}_{n=1}^{\infty}$ be a deterministic sequence of training datasets where $\mathbf{Z}^{(n)}$ has size n . Define

$$\hat{f}_{n, \infty}(x; \mathbf{Z}^{(n)}) = \mathbb{E}_{s \sim \mathcal{P}_{\mathcal{S}}} \left[\hat{f}_n(x; \mathbf{Z}^{(n)}, s) \right].$$

For any test data $x \in \mathcal{X}$, assume that as $n \rightarrow \infty$

$$\mathbb{E}_{s \sim \mathcal{P}_{\mathcal{S}}} \left[\left(\hat{f}_n(x; \mathbf{Z}^{(n)}, s) - \hat{f}_{n, \infty}(x; \mathbf{Z}^{(n)}) \right)^2 \right] \rightarrow 0.$$

3. Subbagging

- Introduce what bagging is
- Touch on all the variants
- Discuss how it has algorithmic stability guarantees
- Write out the subbagging algorithm

Let

$$B_j(m, n) = \{i_1, \dots, i_m\} \subset \{1, \dots, n\}, \quad \text{where } 1 \leq m \leq n$$

denote a m -size index subsample constructed without replacement. There are then $K = \binom{n}{m}$ possible subsamples; denote the collection of all the m -size subsamples as $\mathcal{B} = \{B_j(m, n)\}_{j=1}^K$; define $\mathcal{P}_{\mathcal{B}} = \text{Unif}(\mathcal{B})$. In a slight abuse of notation, let $\mathbf{Z}(B_j)$ be the sub-dataset consisting of the observations of $\mathbf{Z}$ indexed by $B_j(m, n)$. Further, let $B_j^c(n - m, n) = \{1, \dots, n\} \setminus B_j(m, n)$ denote the index complement of $B_j(m, n)$ so that $\mathbf{Z}(B_j^c)$ is the sub-dataset of observations not in $\mathbf{Z}(B_j)$; in what follows, we will refer to these observations as *out-of-bag*.

4. Bagging controls random seed stability

Assumption 2. Let $\{\mathbf{Z}^{(n)}\}_{n=1}^{\infty}$ be a deterministic sequence of training datasets where $\mathbf{Z}^{(n)}$ has size n , and $\rho \in (0, 1)$ is a fixed constant. Define $\mathcal{B}^{(n)} = \{B_j(m, n)\}_{j=1}^K$ as the set of all K possible m -sized subsets of $\mathbf{Z}^{(n)}$ where $m = \lfloor \rho n \rfloor$. Define

$$\bar{f}_{m,\infty}(x; \mathcal{B}) = \frac{1}{K} \sum_{j=1}^K \mathbb{E}_{s \sim \mathcal{P}_S} [\hat{f}_m(x; \mathbf{Z}(B_j), s)]$$

as the idealized subbagged estimator. For any test data $x \in \mathcal{X}$, assume as $n \rightarrow \infty$

$$\mathbb{E}_{\zeta \sim \mathcal{P}_S \otimes \mathcal{P}_{\mathcal{B}}} \left[\left(\bar{f}_m(x; \mathcal{A}^{(m)}, \sigma) - \bar{f}_{m,\infty}(x; \mathcal{B}^{(n)}) \right)^2 \right] \rightarrow 0,$$

where $\zeta = \{(s_1, B_1), \dots, (s_V, B_V)\}$ is the set of V independent random seed-and-bag pairs, each drawn from the product measure $\mathcal{P}_S \otimes \mathcal{P}_{\mathcal{B}}$.

Theorem 4.1. Assume $\mathbb{E}_{\zeta \sim \mathcal{P}_S \otimes \mathcal{P}_{\mathcal{B}}} \left[\left(\bar{f}_m(x; \mathcal{A}, \sigma) - \bar{f}_{m,\infty}(x; \mathcal{B}^{(n)}) \right)^2 \right]$ decays at rate $\frac{\sigma^2}{Vg(\rho n)}$. Then $\dots$ is (ϵ, δ) -random seed stable whenever $V \geq \dots$

4.1. Numerical experiments

5. Relevance to causal machine learning

5.1. The out-of-bag Super Learner

5.2. Cross-bagging

6. Discussion

Proofs

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Supplementary Material

Simulation R code

Short description of Supplement A.